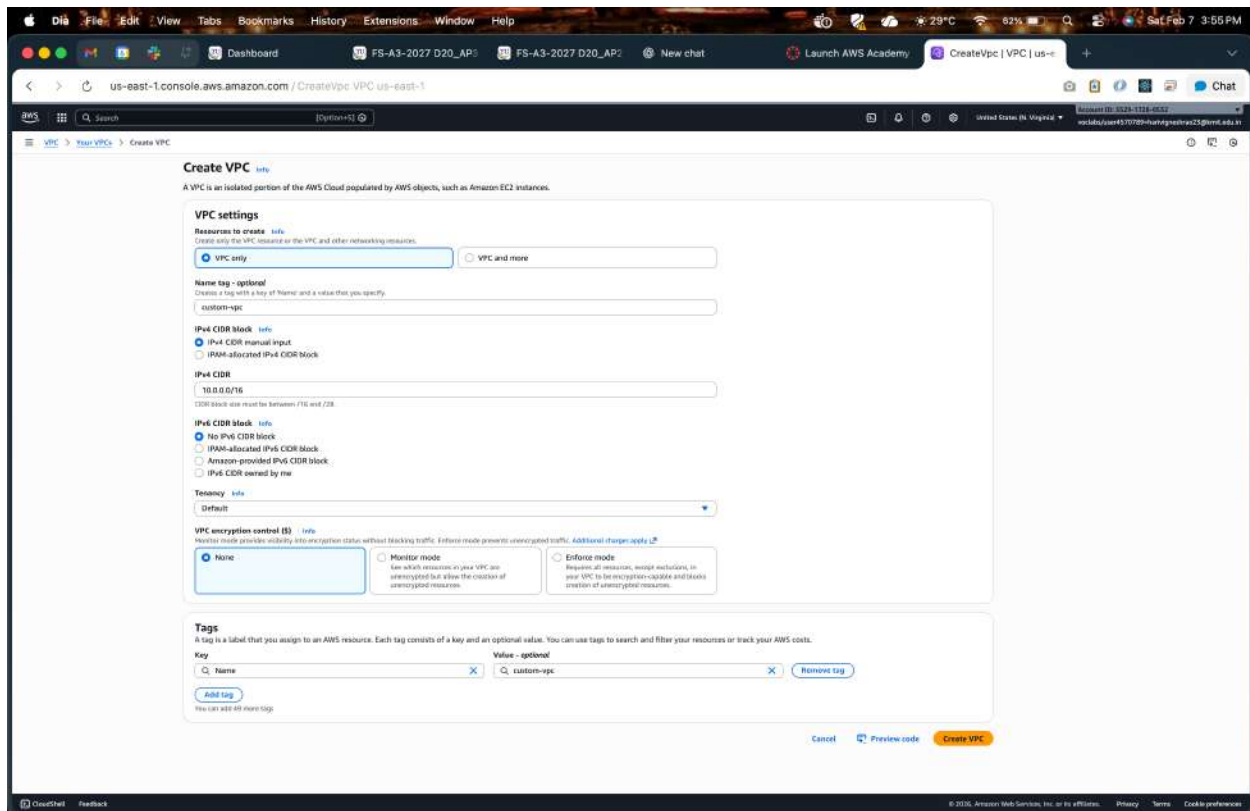


Week 5: VPC Creation and Ec2 Instance Connection

HARI VIGNESH RAO 23BD1A052Q

◆ Step 1: Create the VPC

1. Go to **AWS Management Console** → **VPC**
2. Click **Create VPC**
3. Choose **VPC only**
4. Enter:
 - **Name:** Custom-VPC
 - **IPv4 CIDR block:** e.g. 10.0.0.0/16
5. Click **Create VPC**



◆ Step 2: Create Subnets (Public & Private)

Create subnets in same **Availability Zones** for high availability.

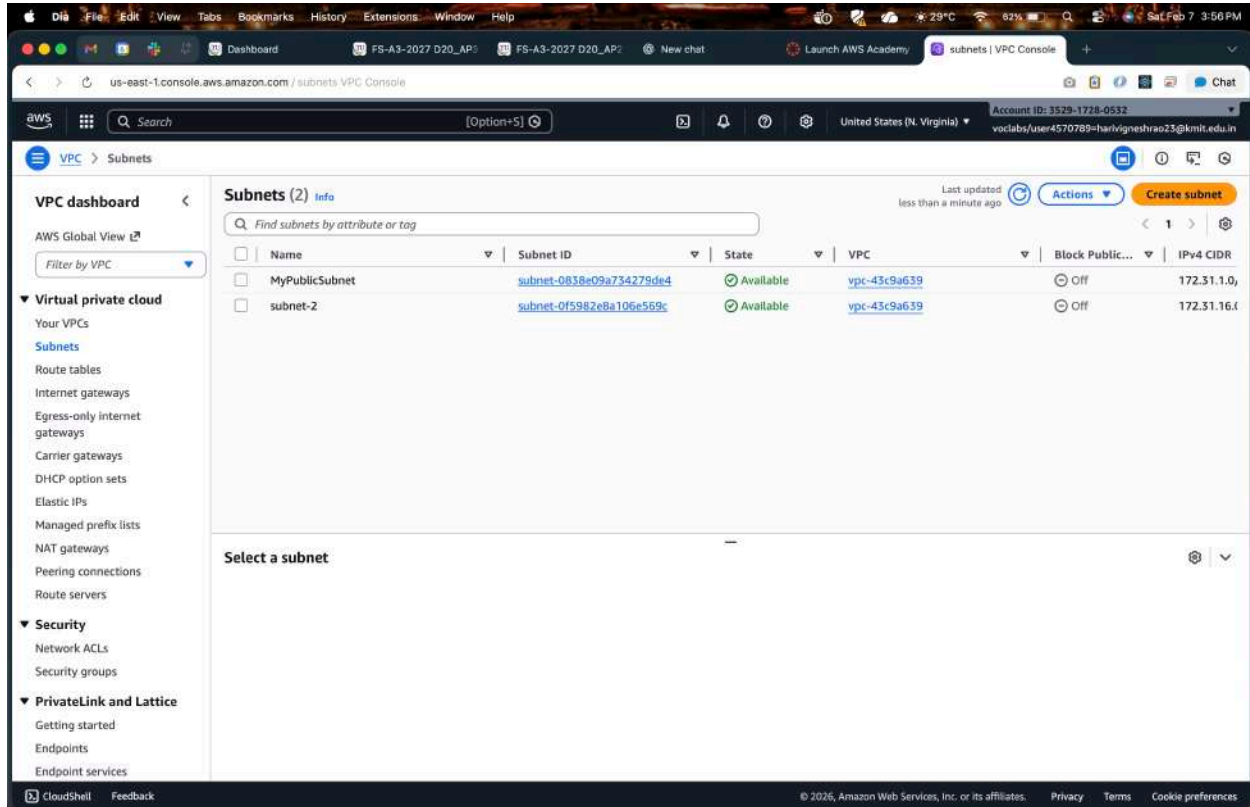
Example:

- **Public Subnet:** 10.0.1.0/24 (AZ-A)

- **Private Subnet:** 10.0.2.0/24 (AZ-A)

Steps:

1. Go to **Subnets** → **Create subnet**
2. Select your VPC
3. Choose AZ
4. Enter CIDR
5. Create subnet



us-east-1 console.aws.amazon.com / VPC us-east-1

Create subnet

VPC

VPC ID
Create subnets in this VPC:
vpc-05b440531806cee8 (custom-vpc)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
public subnet
The name can be up to 256 characters long.

Availability Zone
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.1.0/24

Tags - optional

Key	Value - optional	
Name	public subnet	Remove

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

us-east-1 console.aws.amazon.com / VPC us-east-1

Create subnet

Tags - optional

Key	Value - optional	
Name	public subnet	Remove

[Add new tag](#)
You can add 40 more tags.
[Remove](#)

Subnet 2 of 2

Subnet name
Create a tag with a key of 'Name' and a value that you specify.
private subnet
The name can be up to 256 characters long.

Availability Zone
Choose the zone in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1a

IPv4 VPC CIDR block
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24

Tags - optional

Key	Value - optional	
Name	private subnet	Remove

[Add new tag](#)
You can add 40 more tags.
[Remove](#)

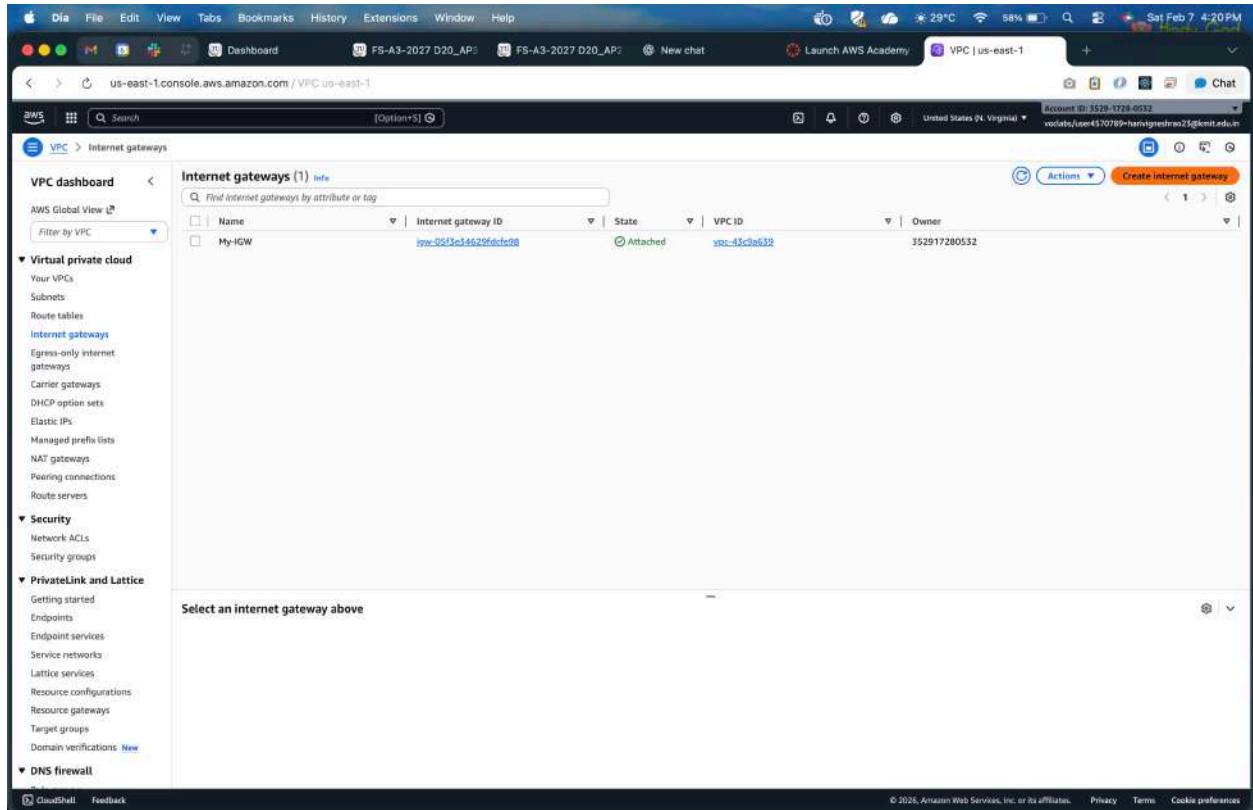
[Add new subnet](#)

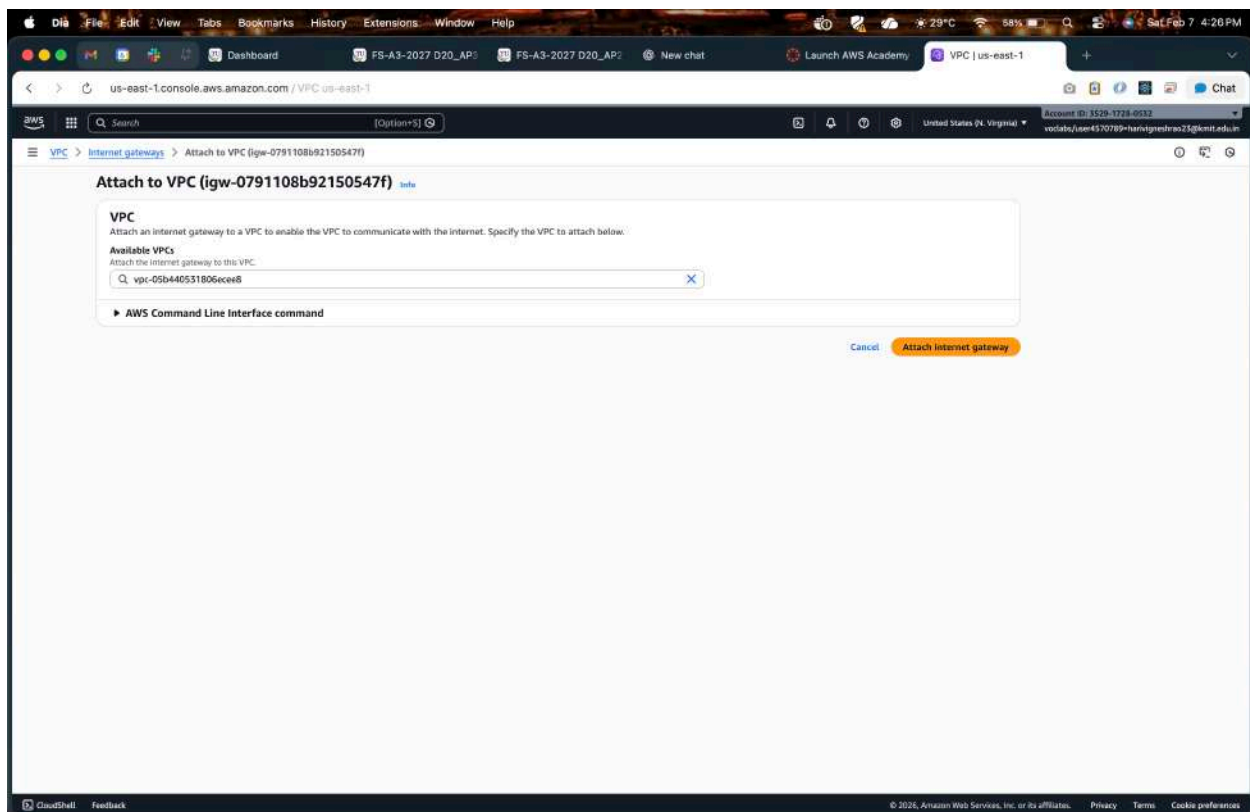
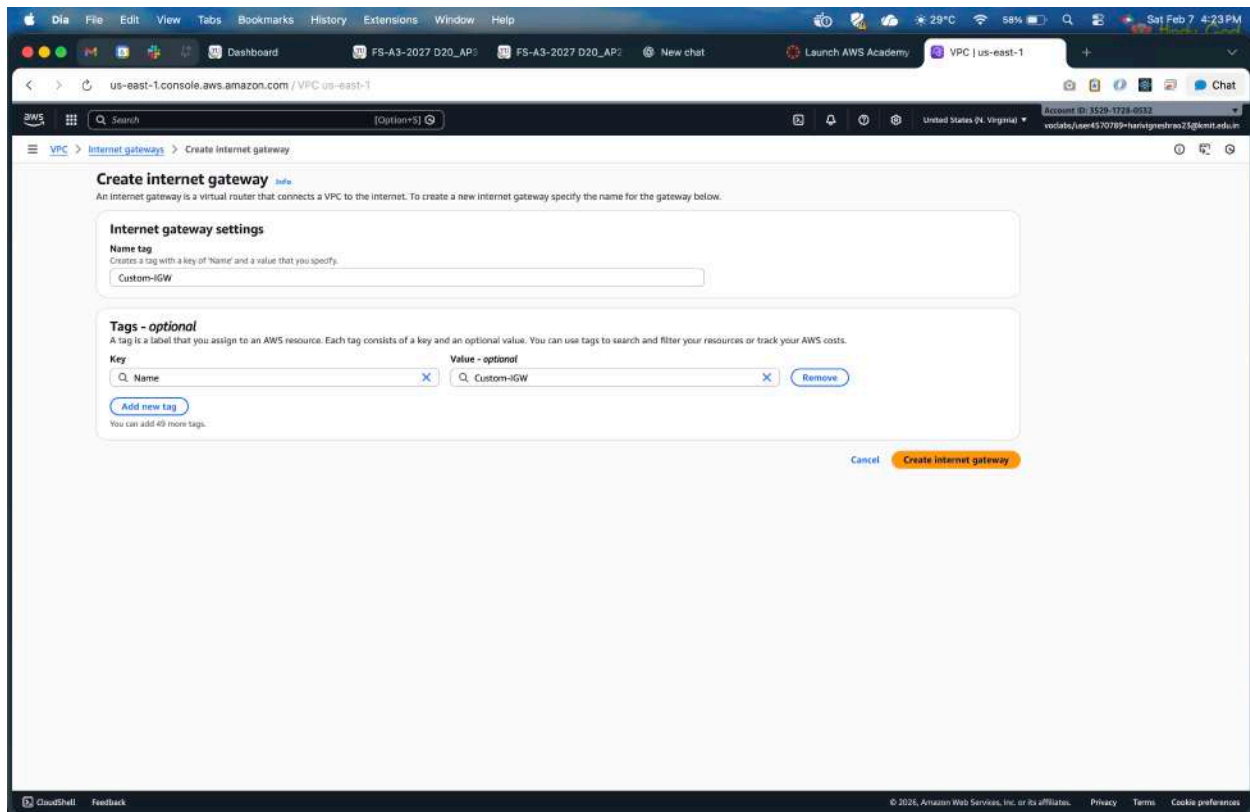
[Cancel](#) [Create subnet](#)

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

◆ Step 3: Create an Internet Gateway (IGW)

1. Go to **Internet Gateways**
2. Click **Create internet gateway**
3. Name it `Custom-IGW`
4. **Attach** it to your VPC





◆ Step 4: Create Route Tables

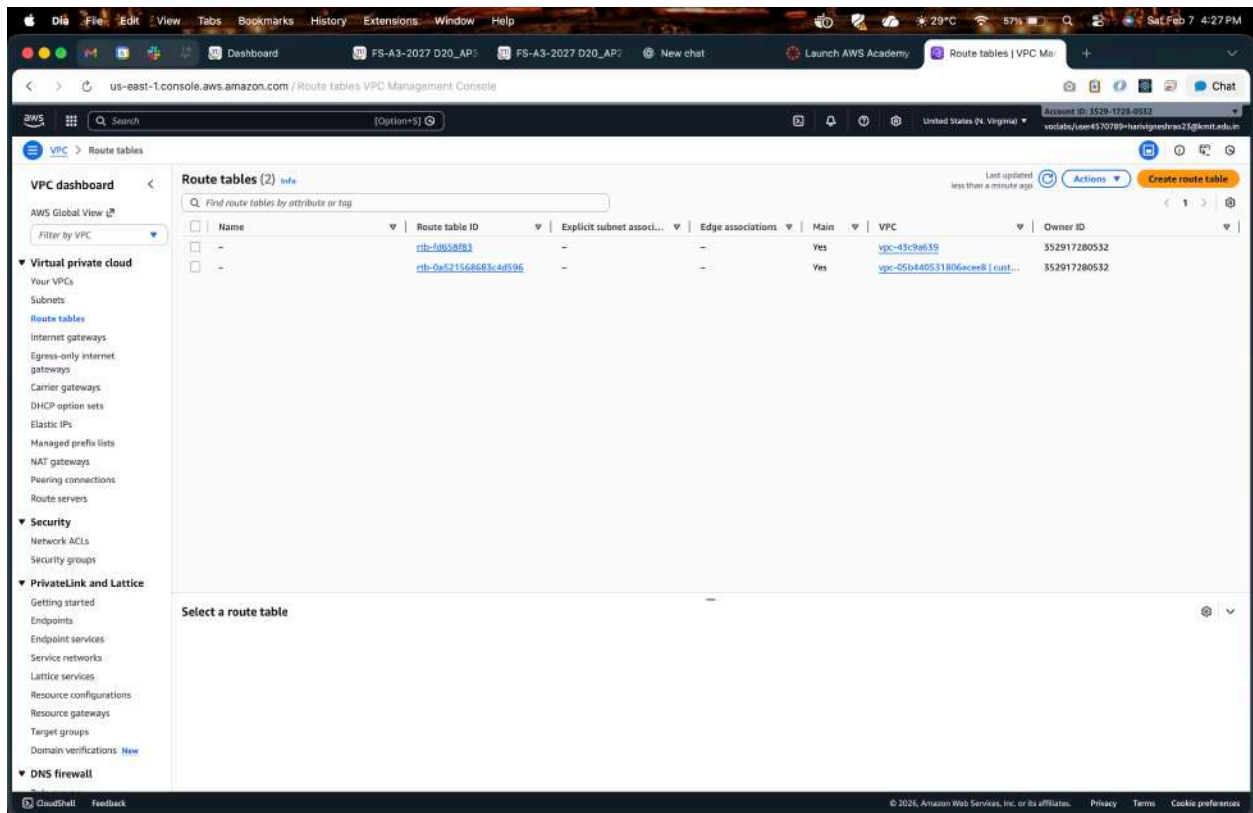
You need **separate route tables** for public and private subnets.

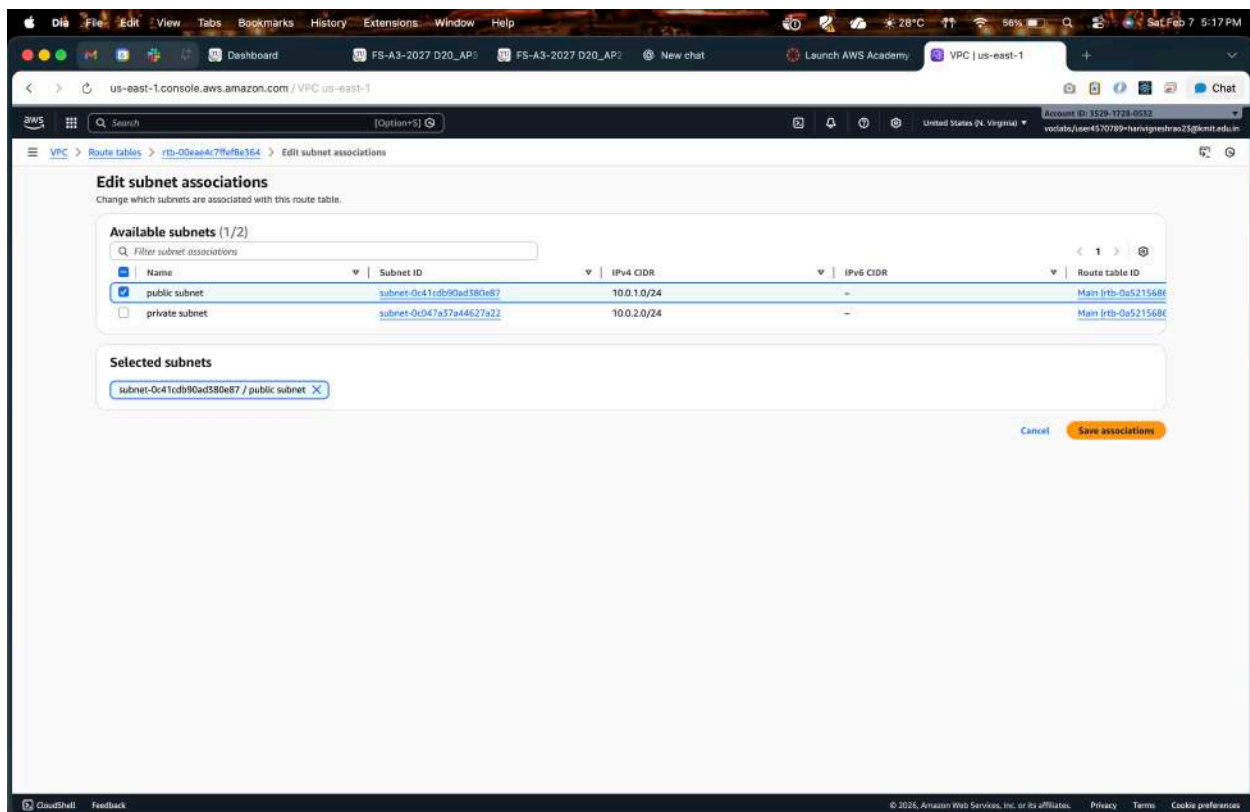
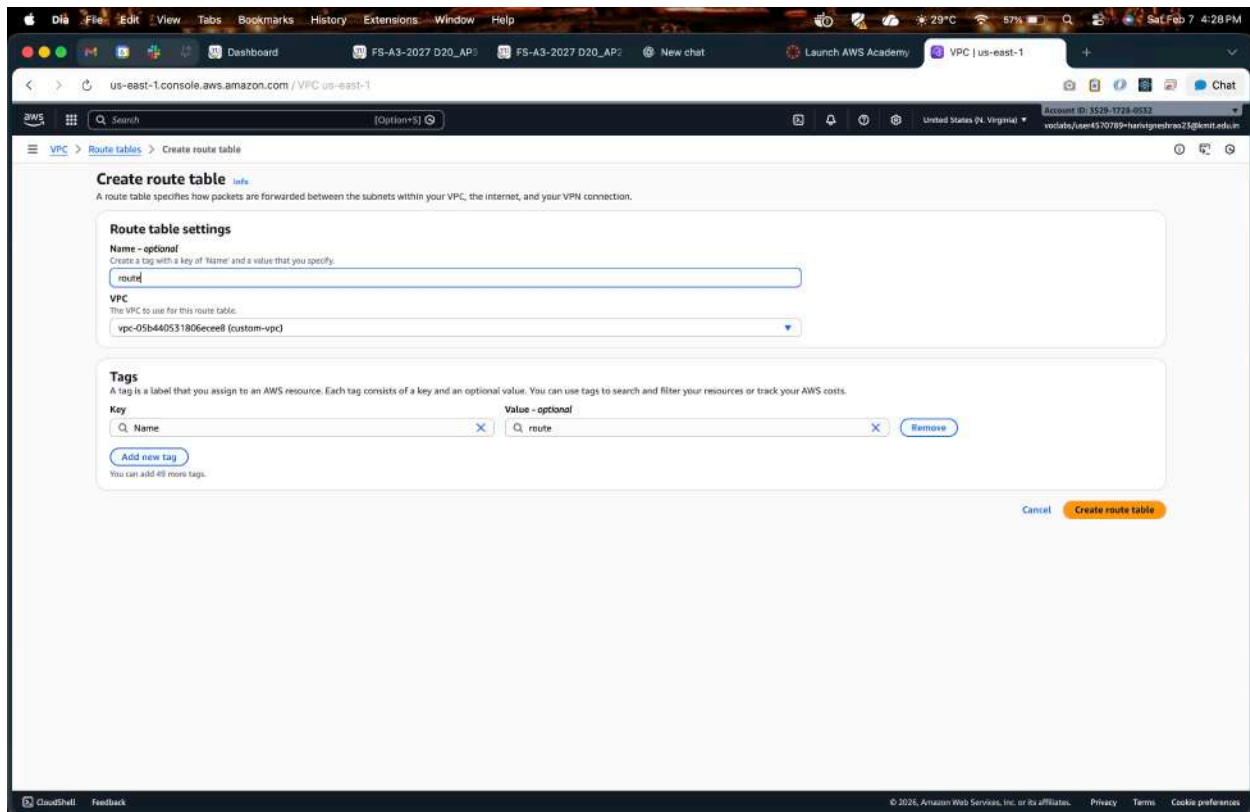
Public Route Table

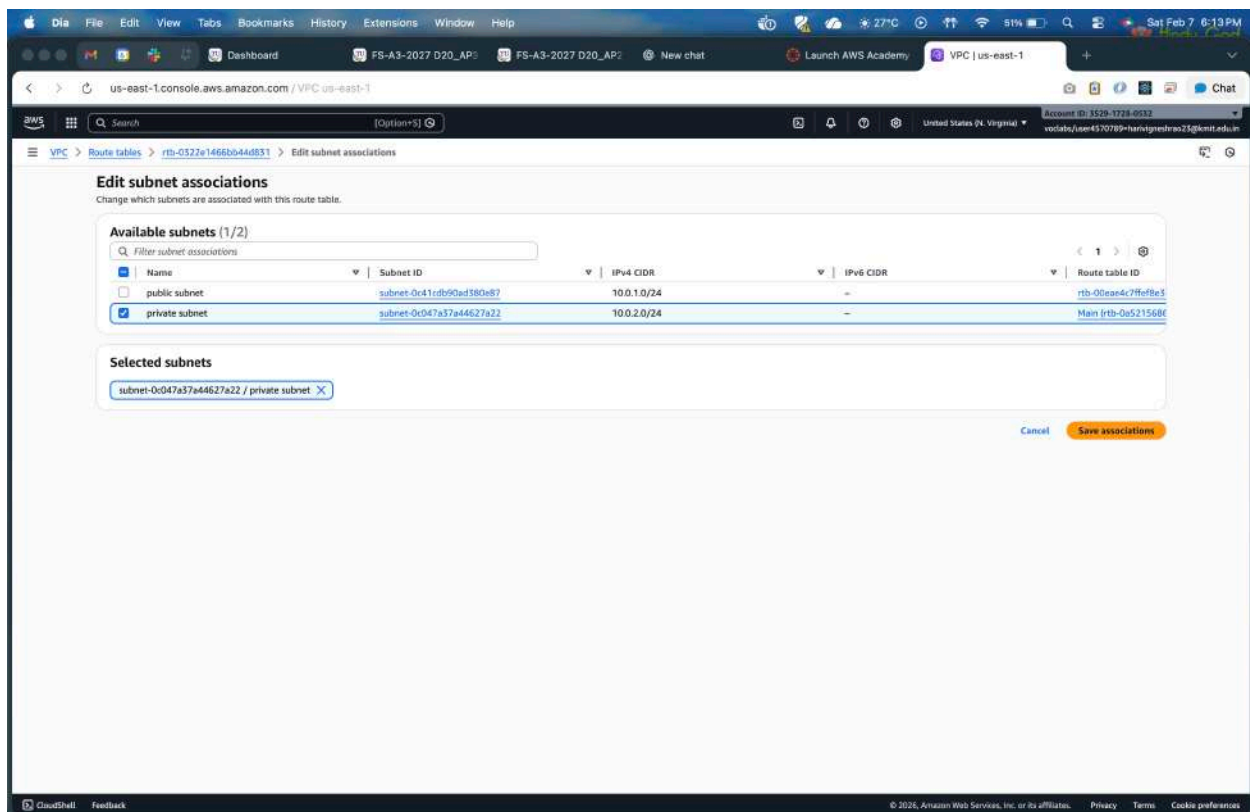
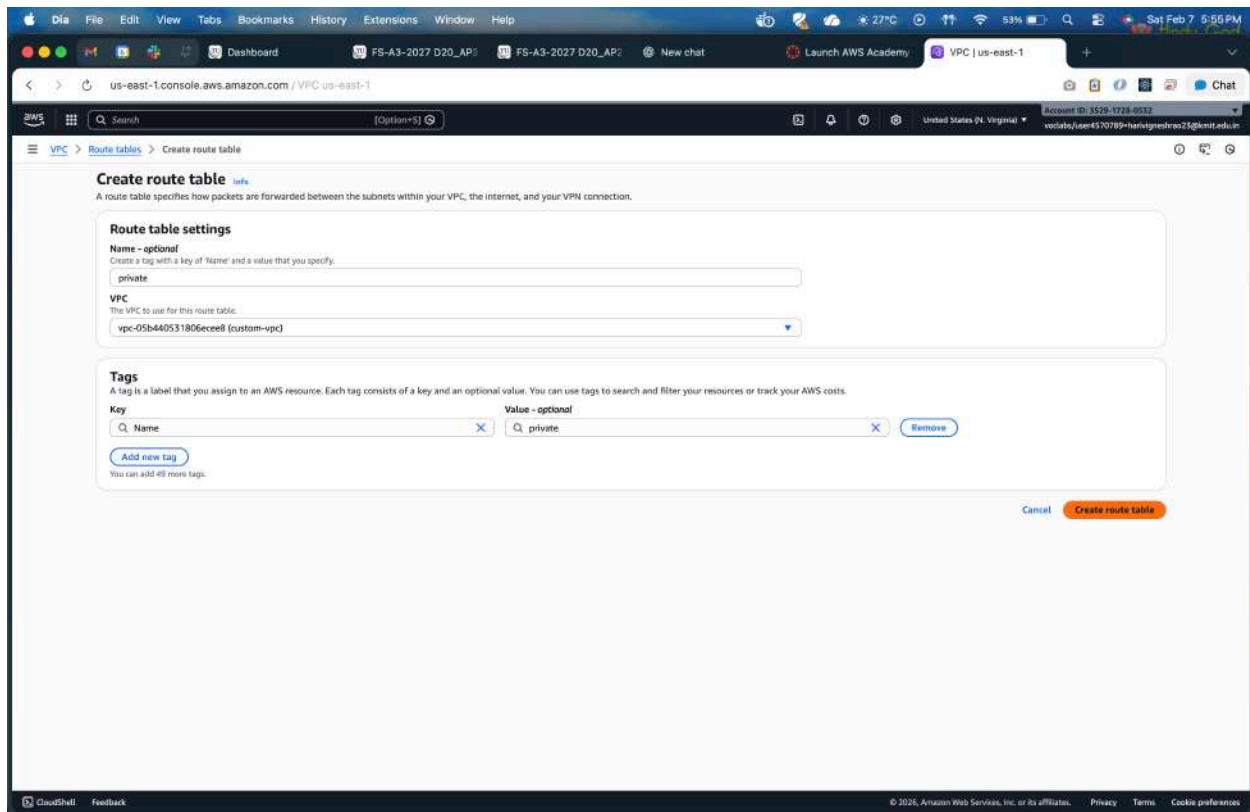
1. Go to **Route Tables** → **Create route table**
2. Select VPC
3. Add route:
 - Destination: 0.0.0.0/0
 - Target: **Internet Gateway**
4. Associate with **public subnet**

Private Route Table

- Keep default local route only (no IGW)
- Associate with **private subnet**

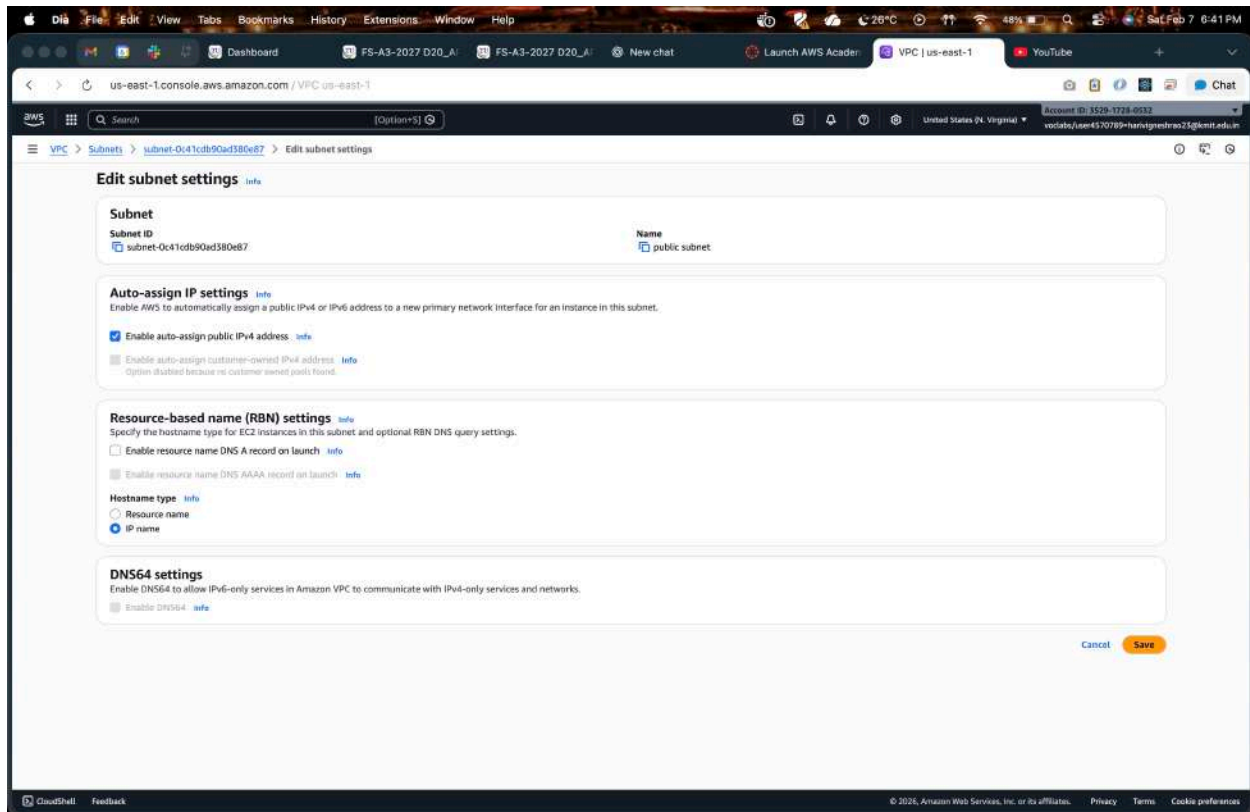






Step 5: Enable Auto-Assign Public IP (Public Subnet)

1. Select **Public Subnet**
2. Go to **Edit subnet settings**
3. Enable **Auto-assign public IPv4 address**



◆ Step 6: Configure Security Groups

1. Create a **Security Group**
2. Add inbound rules:
 - HTTP (80) / HTTPS (443)
 - SSH (22) from trusted IP
3. Attach to EC2 instances

us-east-1.console.aws.amazon.com / VPC Console

VPC > Security Groups

VPC dashboard

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only Internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

PrivateLink and Lattice

Getting started

Endpoints

Endpoint services

Service networks

Lattice services

Resource configurations

Resource gateways

Target groups

Domain verifications

DNS firewall

Security Groups (21)

Find security groups by attribute or tag

Name	Security group ID	Security group name	VPC ID	Description	Owner
launch-wizard-6	sg-0f50d620d89b2d4dc	launch-wizard-6	vpc-43c9a639	launch-wizard-6 created 2026-01-09T0...	352917280532
launch-wizard-2	sg-0f626c07c5c72949	launch-wizard-2	vpc-43c9a639	launch-wizard-2 created 2026-01-09T0...	352917280532
launch-wizard-9	sg-0173b13a12a987111	launch-wizard-9	vpc-43c9a639	launch-wizard-9 created 2026-01-17T0...	352917280532
launch-wizard-5	sg-0e098a6d3ad2c74db	launch-wizard-5	vpc-43c9a639	launch-wizard-5 created 2026-01-09T0...	352917280532
launch-wizard-7	sg-0c27b79a63b411bdf	launch-wizard-7	vpc-43c9a639	launch-wizard-7 created 2026-01-09T1...	352917280532
launch-wizard-15	sg-09c4f6a538b08552	launch-wizard-15	vpc-43c9a639	launch-wizard-15 created 2026-01-24T...	352917280532
launch-wizard-3	sg-04374b811b54074cf	launch-wizard-3	vpc-43c9a639	launch-wizard-3 created 2026-01-09T0...	352917280532
launch-wizard-16	sg-09edc1a8a05f703d4	launch-wizard-16	vpc-43c9a639	launch-wizard-16 created 2026-01-24T...	352917280532
launch-wizard-8	sg-02ba8835229bfc722	launch-wizard-8	vpc-43c9a639	launch-wizard-8 created 2026-01-09T1...	352917280532
launch-wizard-12	sg-0666510aca2797bec	launch-wizard-12	vpc-43c9a639	launch-wizard-12 created 2026-01-24T...	352917280532
launch-wizard-17	sg-02a92724628ca0f8	launch-wizard-17	vpc-43c9a639	launch-wizard-17 created 2026-01-24T...	352917280532
default	sg-041576606526b001a	default	vpc-05b440531806cece8	default VPC security group	352917280532
launch-wizard-4	sg-0d35c75c858c7c45d	launch-wizard-4	vpc-43c9a639	launch-wizard-4 created 2026-01-09T0...	352917280532
launch-wizard-11	sg-03a40490c5d51957a	launch-wizard-11	vpc-43c9a639	launch-wizard-11 created 2026-01-24T...	352917280532
launch-wizard-14	sg-088933fc03a71a1c1	launch-wizard-14	vpc-43c9a639	launch-wizard-14 created 2026-01-24T...	352917280532
default	sg-033a1b0b3	default	vpc-43c9a639	default VPC security group	352917280532
launch-wizard-13	sg-0a234d8a4c2d8cb9	launch-wizard-13	vpc-43c9a639	launch-wizard-13 created 2026-01-24T...	352917280532
launch-wizard-10	sg-025da5e47a13ee7f4	launch-wizard-10	vpc-43c9a639	launch-wizard-10 created 2026-01-24T...	352917280532

Select a security group

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

us-east-1.console.aws.amazon.com / VPC us-east-1

VPC > Security Groups > Create security group

Basic details

Security group name info

vpccorety

Name cannot be edited after creation.

Description info

allows ssh to developers

VPC info

vpc-43c9a639

Inbound rules

Type	Protocol	Port range	Source	Description - optional
HTTP	TCP	80	Anywhere...	0.0.0.0/0
HTTPS	TCP	443	Anywhere...	0.0.0.0/0
SSH	TCP	22	My IP	183.82.238.18/32

Add rule

Rules with source of 0.0.0.0/0 or ::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules

Type	Protocol	Port range	Destination	Description - optional
All traffic	All	All	Custom	0.0.0.0/0

Add rule

Rules with destination of 0.0.0.0/0 or ::0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

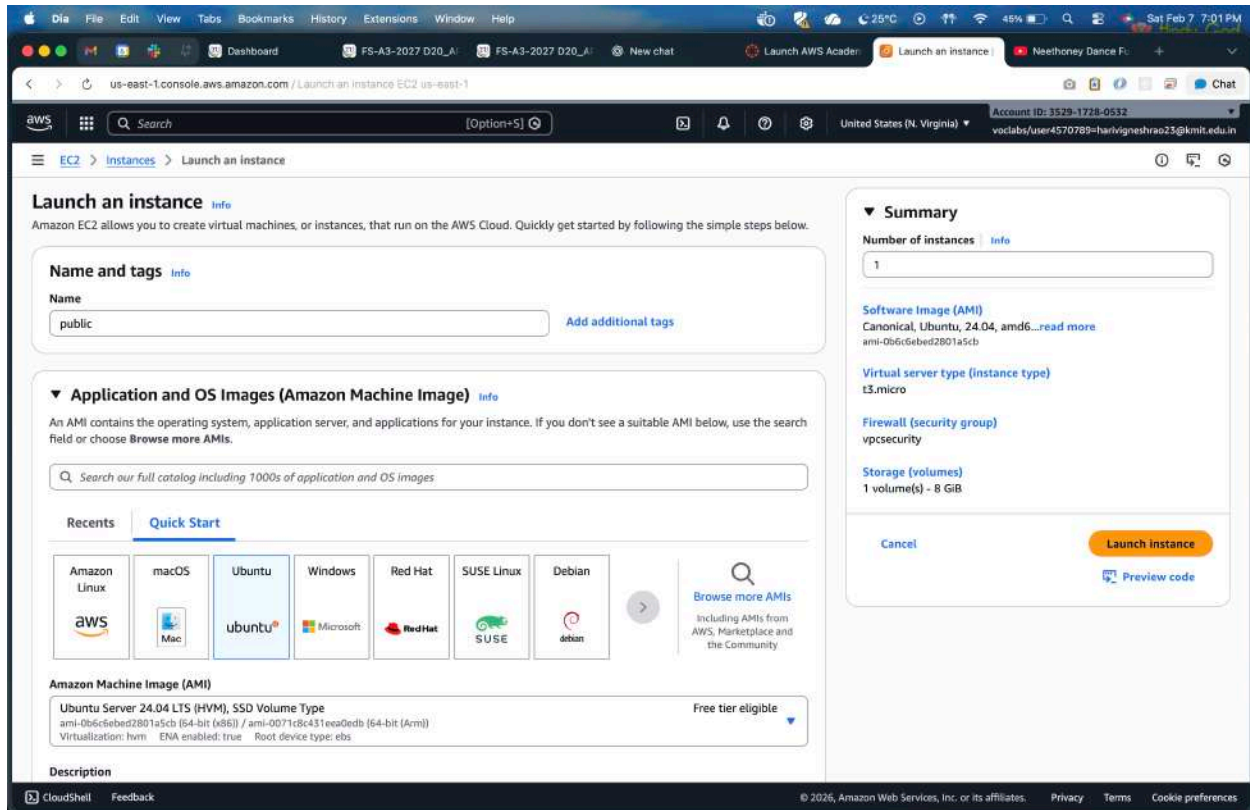
Add new tag

You can add up to 50 more tags

© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

◆ Step 7: Launch EC2 Instances

- Public EC2 → Public subnet
 - Private EC2 → Private subnet
- Attach correct security groups.



us-east-1 console.aws.amazon.com / Launch an instance EC2 us-east-1

EC2 > Instances > Launch an instance

Network settings

VPC - required

vpc-43c9a639 (default)

Subnet

subnet-0838e09a734279de4 MyPublicSubnet

Auto-assign public IP

Enable

Firewall (security groups)

Create security group Select existing security group

Common security groups

Select security groups

vpcsecurity sg-010cadf51f52f867

Advanced network configuration

Configure storage

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Summary

Number of instances 1

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-0b6c6ebd2801a5cb

Virtual server type (instance type)
t3.micro

Firewall (security group)
vpcsecurity

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

us-east-1 console.aws.amazon.com / Launch an instance EC2 us-east-1

EC2 > Instances > Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

private

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0532be01f26a3de55 (64-bit (x86), uefi-preferred) / ami-0bb7267a511c0a8e8 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Summary

Number of instances 1

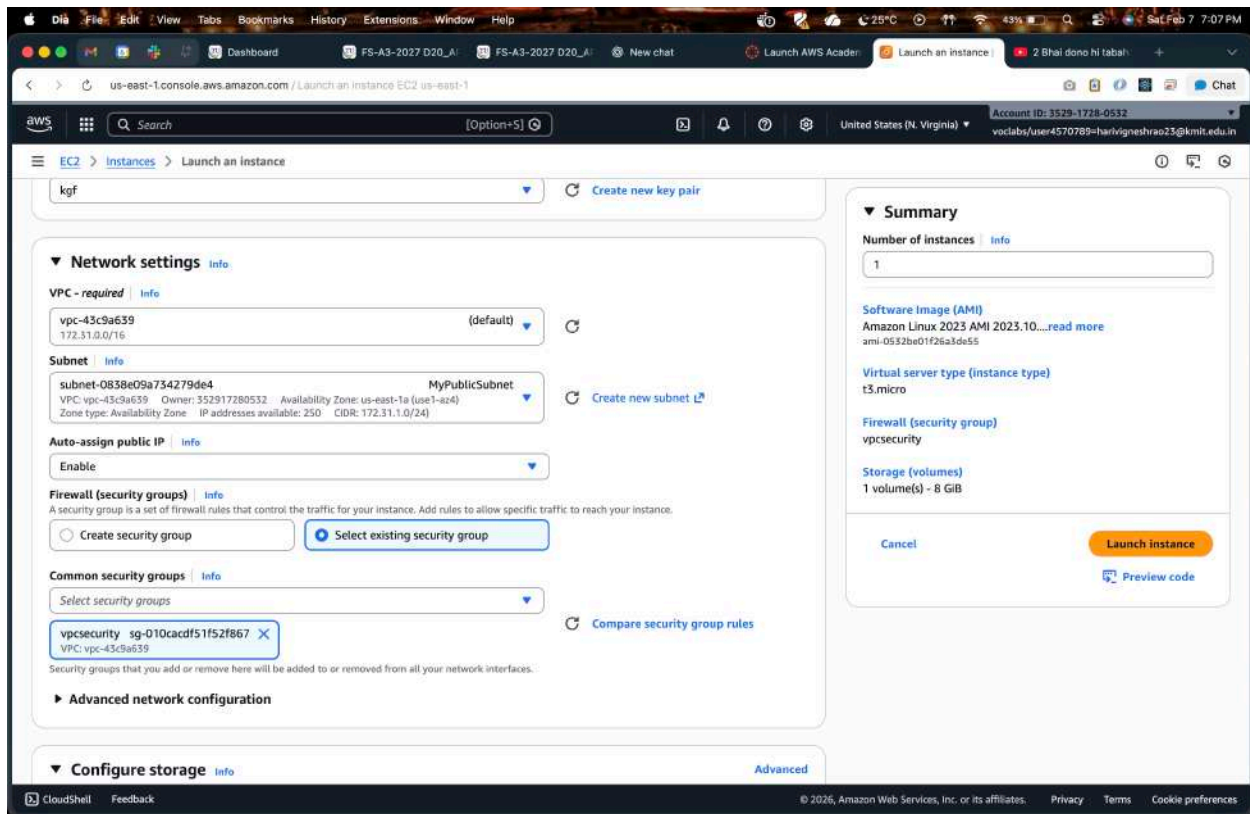
Software Image (AMI)
Amazon Linux 2023 AMI 2023.10...read more
ami-0532be01f26a3de55

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code



◆ Step 8: Test the Setup

- Public EC2 → Internet access ✓


```
Terminal Shell Edit View Window Help
Downloads — ubuntu@ip-172-31-1-76: ~ — ssh -i kgf.pem ubuntu@ec2-107-22-37-4.compute-1.amazonaws.com — 213x60

Last login: Sat Feb 7 09:53:59 on ttyS001
harivigneshrao@Marius-MacBook-Pro: ~ % cd Downloads
harivigneshrao@Marius-MacBook-Pro: Downloads % ssh -i "kgf.pem" ubuntu@ec2-107-22-37-4.compute-1.amazonaws.com
The authenticity of host 'ec2-107-22-37-4.compute-1.amazonaws.com (107.22.37.4)' can't be established.
ED25519 key fingerprint is SHA256:rkfG2Je2IDJqbr10ICxdr79B/X9rA7P+eZU7++f1fHE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-107-22-37-4.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Feb 7 13:35:49 UTC 2020

System load: 0.01          Temperature: -273.1 C
Usage of /: 26.0% of 6.71GB Processes: 115
Memory usage: 25%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.1.76

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-1-76:~$
```

-
- Private EC2 → Internet via NAT ✓

```
Terminal Shell Edit View Window Help
Downloads — ec2-user@ip-172-31-1-40: ~ — ssh -i kgf.pem ec2-user@ec2-54-234-17-175.compute-1.amazonaws.com — 213x60

ED25519 key fingerprint is SHA256:rkfG2Je2IDJqbr10ICxdr79B/X9rA7P+eZU7++f1fHE.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-107-22-37-4.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.14.0-1018-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Feb 7 13:35:49 UTC 2020

System load: 0.01          Temperature: -273.1 C
Usage of /: 26.0% of 6.71GB Processes: 115
Memory usage: 25%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.1.76

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-1-76:~$ ^C
ubuntu@ip-172-31-1-76:~$ exit
logout
Connection to ec2-107-22-37-4.compute-1.amazonaws.com closed.
harivigneshrao@Marius-MacBook-Pro: Downloads % ssh -i "kgf.pem" ec2-user@ec2-54-234-17-175.compute-1.amazonaws.com
The authenticity of host 'ec2-54-234-17-175.compute-1.amazonaws.com (54.234.17.175)' can't be established.
ED25519 key fingerprint is SHA256:8KYWQ2Uq1K8s9r1pCJQ6+EXXK5ExpJ190YlywP740.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-234-17-175.compute-1.amazonaws.com' (ED25519) to the list of known hosts.

#_
_###_      Amazon Linux 2023
_####_
_###_
_#/_
_#/_      https://aws.amazon.com/linux/amazon-linux-2023
_#/_
_#/_
_#/_

[ec2-user@ip-172-31-1-40 ~]$
```

-
- Private EC2 → No direct inbound access ✓

